MILITARY SPECIFICATION

ADHESIVE, CONTACT, FOR CUSTOM FIT HELMET LINERS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a flexible adhesive in a volatile organic solvent for contact bonding of leather to polyurethane foam custom fit liners in pilots helmets. It will also bond other materials such as plastic decorative laminates, fabrics, and wood.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-508	Plastic Sheet, Laminated Decorative, and Nondecorative
TT-E-485	Enamel Semi-gloss, Rust-Inhibiting
PPP-C-96	Can, Metal, 28 Gage and Lighter
PPP-D-729	Drum, Shipping and Storage, Steel 55 Gallon

STANDARDS

FEDERAL

FED-STD-123	Marking for Shipment (Civil Agencies)
FED-STD-141	Paint, Varnish, Lacquer and Related Materials, Methods of Inspection, Sampling and Testing

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to the Air Force Materials Laboratory, MXA, WPAFB, Ohio, 45433 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8040

Source: https://assist.dla.mil -- Downloaded: 202640-23T11:51Z
Check the source to verify that this is the current version before use.

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

(Copies of specifications and standards, required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D86	Distillation of Petroleum Products
ASTM D244	Testing Emulsified Asphalts
ASTM D297	Rubber Products - Chemical Analysis
ASTM D1002	Test for Strength Properties of Adhesives in Shear by Tension Loading (Metal to Metal)
ASTM D1084	Test For Consistency of Adhesives
ASTM D1183	Test For Resistance of Adhesive to Cyclic Laboratory Aging Conditions (Tentative)
ASTM D1489	Test For Nonvolatile Content Of Aqueous Adhesives
ASTM D1875	Test for Density of Adhesives in Fluid Form (Tentative)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:51Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Material. The adhesive shall be composed of polychloroprene (neoprene) rubber and synthetic resin modified to the extent that the requirements of this specification are met (see 4.1.1).

3.1.1 Solvents. The solvents used shall not attack polyether based, freon flown, rigid polyurethane foam. Ketones shall not be used. The solvent, when tested as specified in 4.4.2.2, 4.4.2.3 and 4.4.2.4 shall not contain ketones, halogenated compounds, or benzene.

3.2 Nonvolatile content. When tested as specified in 4.4.4, the nonvolatile content shall be in the range of 18 to 25 percent by weight.

3.3 Ash content. When tested as specified in 4.4.4, the ash content of the dry film of the adhesive shall be not less than 3.0 percent or more than 20 percent.

3.4 Viscosity. The viscosity of the adhesive shall be not less than 400 or more than 1500 centipoises when tested as specified in 4.4.5.

3.5 Density. The density in pounds per gallon shall be in the range of 6.5 - 7.3.

3.6 Shear strength. When tested as specified in 4.4.7, the shear strength of the adhesive in pounds per square inch shall be as follows:

	Minimum
Before aging	150
After aging	200

3.7 Bonding range. When tested as specified in 4.4.8, the bonding range of the adhesive shall be a minimum on one hour.

3.8 Accelerated aging. When tested as specified in 4.4.9, the adhesive shall change in viscosity not more than 10 percent plus or minus from the originally specified value (see 3.4), and there shall be no evidence of gelling or separation.

3.9 Shelf storage life. When the adhesive is stored as specified in 4.4.10, there shall be no gelling, stringing, or other non-uniformity; the viscosity shall have changed not more than 10 percent plus or minus from the originally specified viscosity (see 3.4); and the strength values shall be not less than those specified in 3.6 and 3.7.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:51Z
Check the source to verify that this is the current version before use.

3.10 Freeze-thaw resistance. When tested as specified in 4.4.11, the adhesive shall remain homogeneous and shall change in viscosity not more than 10 percent plus or minus from the originally specified value (see 3.4).

3.11 Workmanship. The adhesive shall be uniform, even-spreading, and free of dirt, lumps, foreign matter, coarse particles, and all other defects which would adversely affect its usability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. The contractor shall submit to the contracting officer, a certificate of compliance indicating that the adhesive is composed of polychloroprene rubber and synthetic resin per 3.1, that the adhesive complies with the shelf storage requirements as specified in 3.10, and that the adhesive formulation does not contain ketone solvents and will not attack polyurethane foam per 3.1.1.

4.2 Sampling.

4.2.1 Lot. For the purpose of sampling, a lot of the adhesive shall consist of a manufacturer's batch. A batch is defined as the end product of all raw materials mixed, blended, or processed in a single operation.

4.2.2 Sampling for inspection of filled containers. A random sample of filled containers shall be selected in accordance with MIL-STD-105 at inspection level I and acceptable quality level (AQL) equals 2.5 percent defective to verify compliance with this specification regarding fill, closure, and marking and other requirements not involving tests.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:51Z
Check the source to verify that this is the current version before use.

4.3 Inspection.

4.3.1 Inspection of containers. Each sample filled container shall be examined for defects of construction of the container and the closure, for evidence of leakage, and for unsatisfactory markings; each filled container shall also be gaged to determine the amount to contents. Any container in the sample having one or more defects or under required fill shall be rejected, and if the number of defective containers in any sample exceeds the acceptance number for the appropriate sampling plan of MIL-STD-105, the lot represented by the sample shall be rejected.

4.3.2 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the packaging, packing, and marking comply with the requirements of section 5 of this specification. Defects shall be scored in accordance with Table I. For examination of interior packaging, the sample unit shall be one shipping container fully prepared for delivery selected at random just prior to the closing operations. Sampling shall be in accordance with MIL-STD-105. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 4.0 defects per hundred units.

TABLE I. Classification of preparation for delivery defects.

Examine	Defects
Markings (exterior and interior)	Omitted; improper size, location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Bulging or distortion of container.

Source: https://assist.dla.mil -- Downloaded: 202640-23T11:51Z
Check the source to verify that this is the currentversion before use.

4.3.3 Testing of the end item. The methods of testing specified in 4.4, shall be followed. For purposes of sampling, the lot shall be expressed in units of gallons of adhesive. The sample unit for testing shall be one container of adhesive, randomly selected from containers in the lot. The adhesive shall be placed in separate clean, dry metal or glass containers, sealed, marked, and forwarded to the testing laboratories. The sample size shall be as follows:

Lot size (gallon)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and more	5

Tests shall be performed on a sample unit basis. All test reports shall contain the individual values utilized in expressing the final result. The lot shall be unacceptable if one or more sample unit fails to meet any test requirement specified.

4.4 Test procedures.

4.4.1 Standard test conditions. Unless otherwise specified, all test specimens shall be prepared and tested in a room having a temperature of 73.5° ±2°F (23.1° ±1.1°C) and a relative humidity of 50 +5 percent.

4.4.2 Solvent analysis.

4.4.2.1 Solvent extraction. The solvent shall be distilled from 100 milliliters of type I adhesive according to ASTM D86.

4.4.2.2 Ketones.

4.4.2.2.1 Reagent. Two grams of 2, 4-dinitrophenylhydrazine +4 milliliters of concentrated sulfuric acid +30 milliliters of methanol (slowly added) +10 milliliters of water.

4.4.2.2.2 Procedure. One milliliter of reagent shall be pipetted into a 20 by 170 millimeters (mm) test tube. Ten drops of distillate shall be added and the test tube shaken for 30 seconds. A yellow precipitate or cloud in the reagent layer indicates the presence of ketones. A blank shall be run using one milliliter of reagent and 10 drops of mineral spirits.

4.4.2.3 Halogenated compounds. The presence of halogenated compounds shall be determined in accordance with method 5132 of FED-STD-141.

Source: https://assist.dla.mil -- Downloaded: 202640-23T11:51Z
Check the source to verify that this is the current version before use.

4.4.2.4 Benzene. When the solvent is tested in accordance with 4.4.2.2.1, a trace benzene peak of not more than 2 percent of the toluene peak will be allowed.

4.4.3 Nonvolatile content. The nonvolatile content of the adhesive shall be determined in accordance with ASTM D1489 to determine compliance with the requirements of 3.2.

4.4.4 Ash content. The ash content of the adhesive shall be determined in accordance with the Reference Ash Method of Fillers Analysis of ASTM D297 to determine compliance with the requirements of 3.3.

4.4.5 Viscosity. The viscosity of the adhesive shall be determined in accordance with method B of ASTM D1084 to determine compliance with the requirements of 3.4. A Brookfield Viscosimeter Model RVT or equal, equipped with a No. 2 spindle and operated at 20 rpm shall be used.

4.4.6 Density. The density of the adhesive shall be determined in accordance with ASTM D1875 to determine compliance with the requirements of 3.5.

4.4.7 Shear strength of bonded joints.

4.4.7.1 Before aging. The shear strength before aging shall be determined in accordance with ASTM D1002 to determine compliance with the requirements of 3.6. Adherends shall be ¼-inch-fir plywood, commercial Grade A, and a high pressure decorative plastic laminate of 0.062 inch nominal thickness conforming to L-P-508.

4.4.7.2 After aging. Specimens prepared as specified in 4.4.7.1 shall be exposed to the aging conditions specified in procedure A of ASTM D1183. After the required conditioning, the exposed specimens shall be subjected to the shear strength tests as specified in 4.4.7.1 to determine compliance with the requirements of 3.6.

4.4.8 Bonding range.

4.4.8.1 Sample preparation. One brush coat of adhesive shall be applied with a two-inch-wide brush to one surface of interior-grade fir plywood, 6 inches by 6 inches by ½-inch, and the adhesive shall be allowed to dry for 20 minutes. A second brush coat of adhesive shall be applied to the plywood, and as soon as possible thereafter a brush coat of adhesive shall be applied to the reverse side of a 6 inch by 6 inch by 0.062 inch piece of high pressure decorative plastic laminate conforming to L-P-508.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:51Z
Check the source to verify that this is the current version before use.

4.4.8.2 Procedure. After one hour drying time for the coated specimen, the adhesive-coated surface of the decorative laminate shall be placed squarely on the coated plywood so that one inch of the laminate extends over the plywood on two sides. The entire laminate surface shall be rolled with a small hand roller not over 3 inches wide using the body pressure to establish the bond. The bond shall be determined by attempting to peel the laminate from the plywood.

4.4.9 Accelerated aging. A wide-mouth, 1-pint container of aluminum or steel shall be filled with the adhesive. The container shall be closed securely and stored for 15 days at 140° ±2°F (60° ±1°C). The container of adhesive shall then be stored at standard conditions (see 4.4.1) for 24 hours. The viscosity of the adhesive shall be determined in accordance with 4.4.5 to determine compliance with the requirement of 3.8. Tests shall be conducted in duplicate and averaged. The test specimen shall be examined for separation and gelling.

4.4.10 Shelf storage life. A one pint container, or an equivalent amount of smaller size containers, of the adhesive shall be stored in the original unopened container or containers for 12 months from the date of manufacture at standard conditions (see 4.4.1) and then shall be tested to determine compliance with the requirements of 3.9.

4.4.11 Freeze-thaw resistance. The freeze-thaw resistance of the adhesive shall be determined in accordance with the freezing test of ASTM D244, and the viscosity of the adhesive shall be determined in accordance with 4.4.5 to determine compliance with the requirement of 3.10.

4.4.12 Compatibility with polyurethane foam. A rectangular piece of polyether based polyurethane foam of a nominal 2 pound/ft^3 density of about 4 inches by 6 inches by 1 inch thick shall be coated with the adhesive. After one hour, the foam shall be visually inspected for evidence of any attack by the adhesive. Wrinkling, pitting or other evidence of solvent attack shall be cause for rejection.

5. PACKAGING

5.1 Preservation-packing. Preservation-packing shall be level A, B, or C, as specified (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:51Z
Check the source to verify that this is the current version before use.

5.1.1 Level A.

5.1.1.1 Cans. Quantities shall be packaged in round 1-pint or 1-quart round cans conforming to type V, class 2 of PPP-C-96. Cans shall have an interior coating which shall neither affect, nor be affected by the adhesive. Exterior plan B coating and side seam stripping of PPP-C-96 shall be required. Containers of one half gallon capacity and above shall be provided with wire handles which shall be galvanized or otherwise protectively coated to resistant corrosion.

5.1.2 Level B.

5.1.2.1 Cans. Quantities shall be packaged in round 1-pint or 1-quart round cans conforming to type V, class 2 of PPP-C-96. Cans shall have an interior coating which shall neither affect, nor be affected by the adhesive. The cans shall have exterior coating, plan A of PPP-C-96.

5.1.3 Level C. Adhesive in the quantities as specified (see 6.2) shall be packaged in accordance with the manufacturer's commercial practice, provided that this insures protection for the adhesive during shipment and provides for safe delivery to its destination.

5.2 Packing.

5.2.1 Levels A and B.

5.2.1.1 Cans. Adhesive packaged as specified (see 6.2) shall be packed for domestic or overseas shipment as specified (see 6.2) in accordance with the appendix of PPP-C-96.

5.2.2 Level C. The adhesive, packaged as specified shall be packed in containers that will assure carrier acceptance and safe delivery at destination in compliance with the applicable carrier rules and regulations.

5.3 Marking. In addition to any special marking required by the contract, intermediate and exterior container and unitized load shall be marked in accordance with MIL-STD-129. When specified (see 6.2), the marking of domestic shipments for civil agencies shall be in accordance with FED-STD-123.

5.3.1 Special marking. In addition to markings specified in 5.3, information shall appear on each unit container and shipping container as follows:

a. Manufacturer's instructions for use (to include thinning directions, if applicable).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:51Z
Check the source to verify that this is the current version before use.

b. Date of manufacture (by month and year, not by code).
c. Date of first reinspection (12 months from date of manufacture).
d. Flash point in degrees Celsius and degrees Fahrenheit.
e. Any special directions for storage or use of the adhesive.

6. NOTES

6.1 Intended use. The adhesive covered by this specification is intended specifically for bonding leather to polyurethane foam used as liners in aircraft pilots helmets. This foam is covered by MIL-P-83379. The requirements of this specification are equivalent to those of MMM-A-130 except that the ketones and other solvents which attack polyurethane foam are prohibited.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Size (1-pint or 1-quart).
c. Quantity required.
d. Level of packaging and packing required (see 5.1 and 5.2).
e. Special marking required (see 5.3).

Custodian:
Air Force - 11

Preparing activity:
Air Force - 11

Project No. 8040-F096

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:51Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-A-87134

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

C16061

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:51Z
Check the source to verify that this is the current version before use.